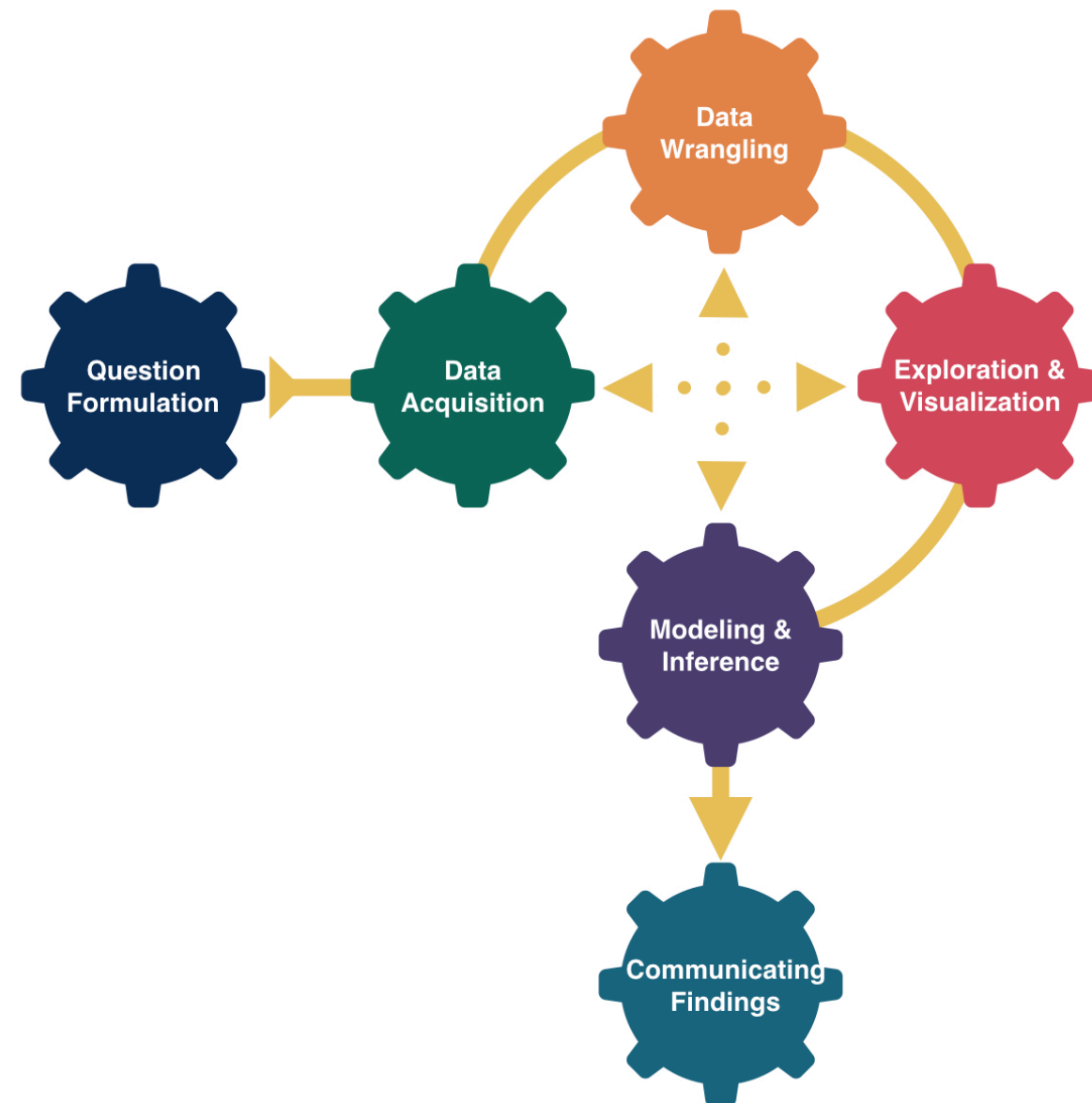


Hypothesis Testing

Kelly McConville

DATA 100

Week 9 | Spring 2026



Announcements

- Map making in **R** workshop: Part of the Sustainability Symposium
 - Th, March 26, 2:30 - 3:50pm, Taylor 210

Goals for Today

- More examples of confidence intervals in **R**
- Confidence interval interpretations
- Set-up the structure of **hypothesis testing**
- Determine if Bucknell students have ESP!

Load Packages and Data

```
1 library(tidyverse)
2 library(infer)
```

Let's return to the movies dataset and estimate numerical quantities about Hollywood movies.

```
1 # Read in data
2 movies <- read_csv("https://www.lock5stat.com/datasets2e/HollywoodMovies.csv")
```

Estimation for a Single Mean

What is the average amount of money (μ) made in the opening weekend?

```
1 # Compute the summary statistic
2 x_bar <- movies %>%
3   drop_na(OpeningWeekend) %>%
4   specify(response = OpeningWeekend) %>%
5   calculate(stat = "mean")
6 x_bar
```

Response: OpeningWeekend (numeric)

A tibble: 1 × 1

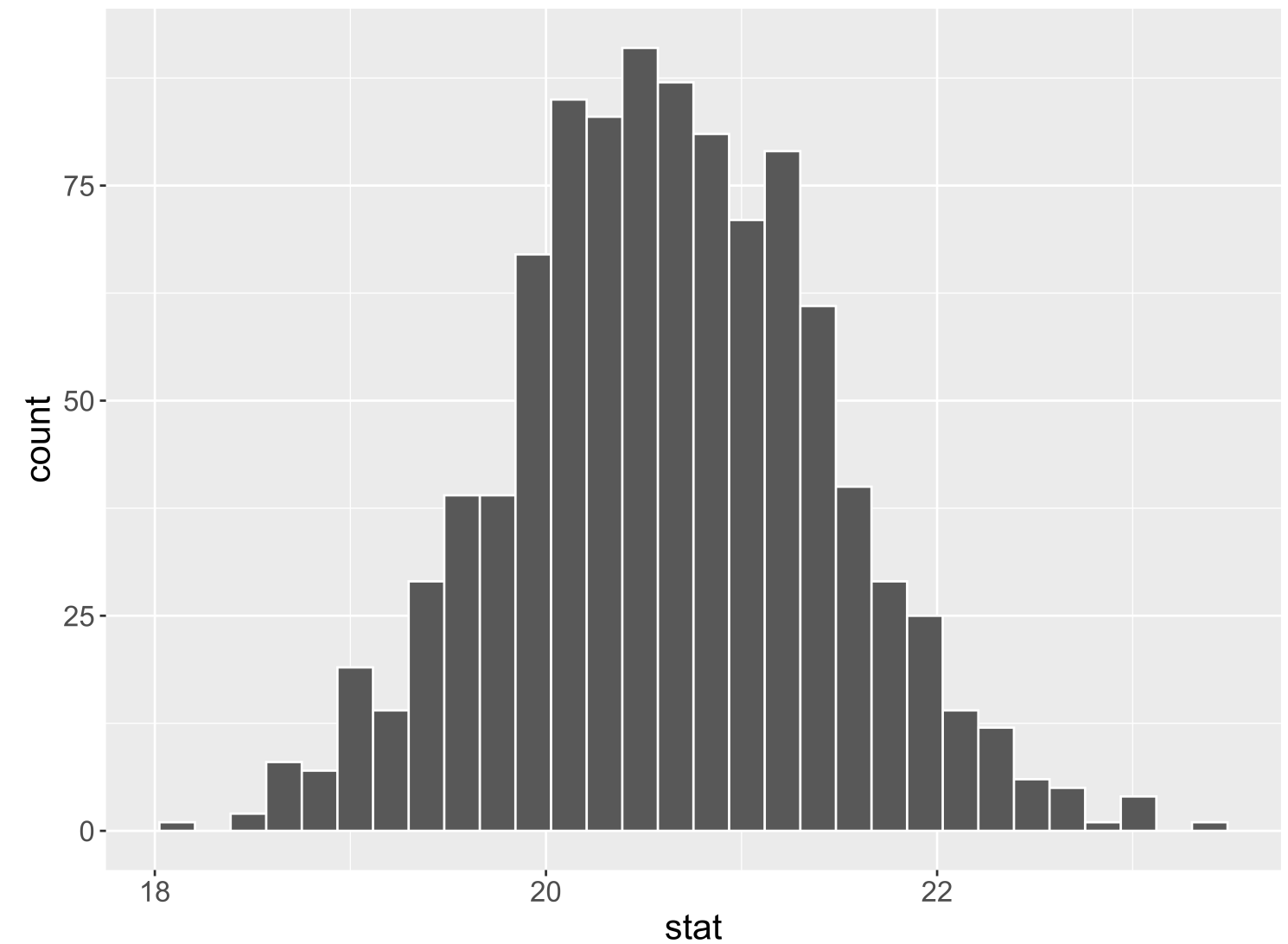
stat

<dbl>

1 20.6

Estimation for a Single Mean

```
1 set.seed(999)
2
3
4 # Construct bootstrap distribution
5 bootstrap_dist <- movies %>%
6   drop_na(OpeningWeekend) %>%
7   specify(response = OpeningWeekend) %>%
8   generate(reps = 1000, type = "bootstrap") %>%
9   calculate(stat = "mean")
10
11 # Look at bootstrap distribution
12 ggplot(data = bootstrap_dist,
13        mapping = aes(x = stat)) +
14   geom_histogram(color = "white")
```



Estimation for a Single Mean – SE Method

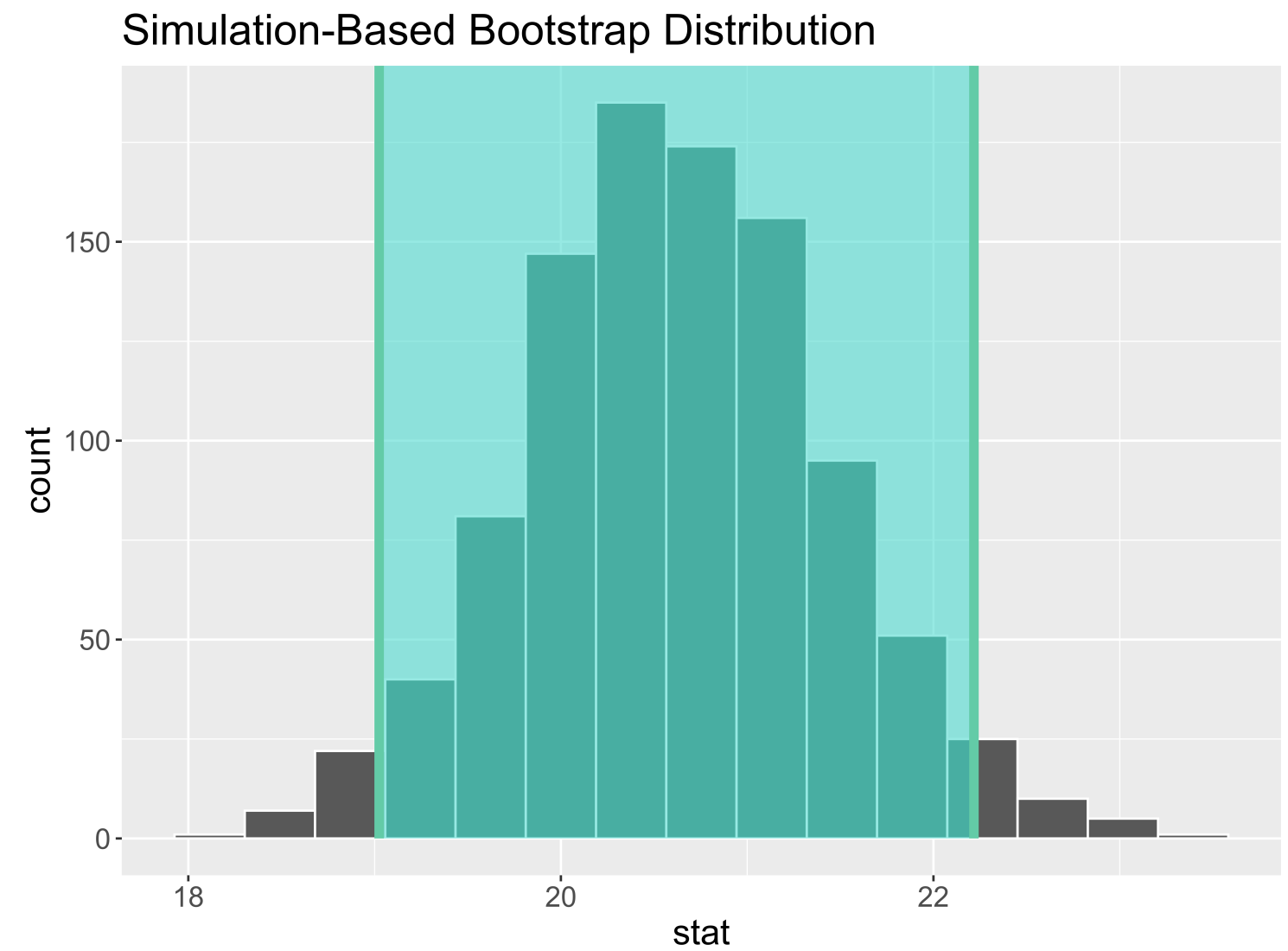
```
1 # Get confidence interval
2 ci <- bootstrap_dist %>%
3   get_confidence_interval(type = "se", level = 0.95,
4                           point_estimate = x_bar)
5 ci
```

```
# A tibble: 1 × 2
  lower_ci upper_ci
  <dbl>    <dbl>
1    19.0    22.2
```

Interpretation: The point estimate is \$ 20.6M. I am 95% confidence that the average amount of money made by all Hollywood movies is between \$ 19.0243721M and \$ 22.2M.

Estimation for a Single Mean

```
1 # Visualize confidence interval
2 bootstrap_dist %>%
3   visualize() +
4   shade_confidence_interval(endpoints = ci)
```



Estimation for a Single Mean – Percentile Method

```
1 # Get confidence interval
2 ci_95 <- bootstrap_dist %>%
3   get_confidence_interval(type = "percentile",
4                           level = 0.95)
5 ci_95
```

```
# A tibble: 1 × 2
  lower_ci upper_ci
  <dbl>     <dbl>
1    19.0     22.3
```

Estimation for Difference in Means

What is the difference in average amount of money made in the opening weekend between action movies and dramas ($\mu_1 - \mu_2$)?

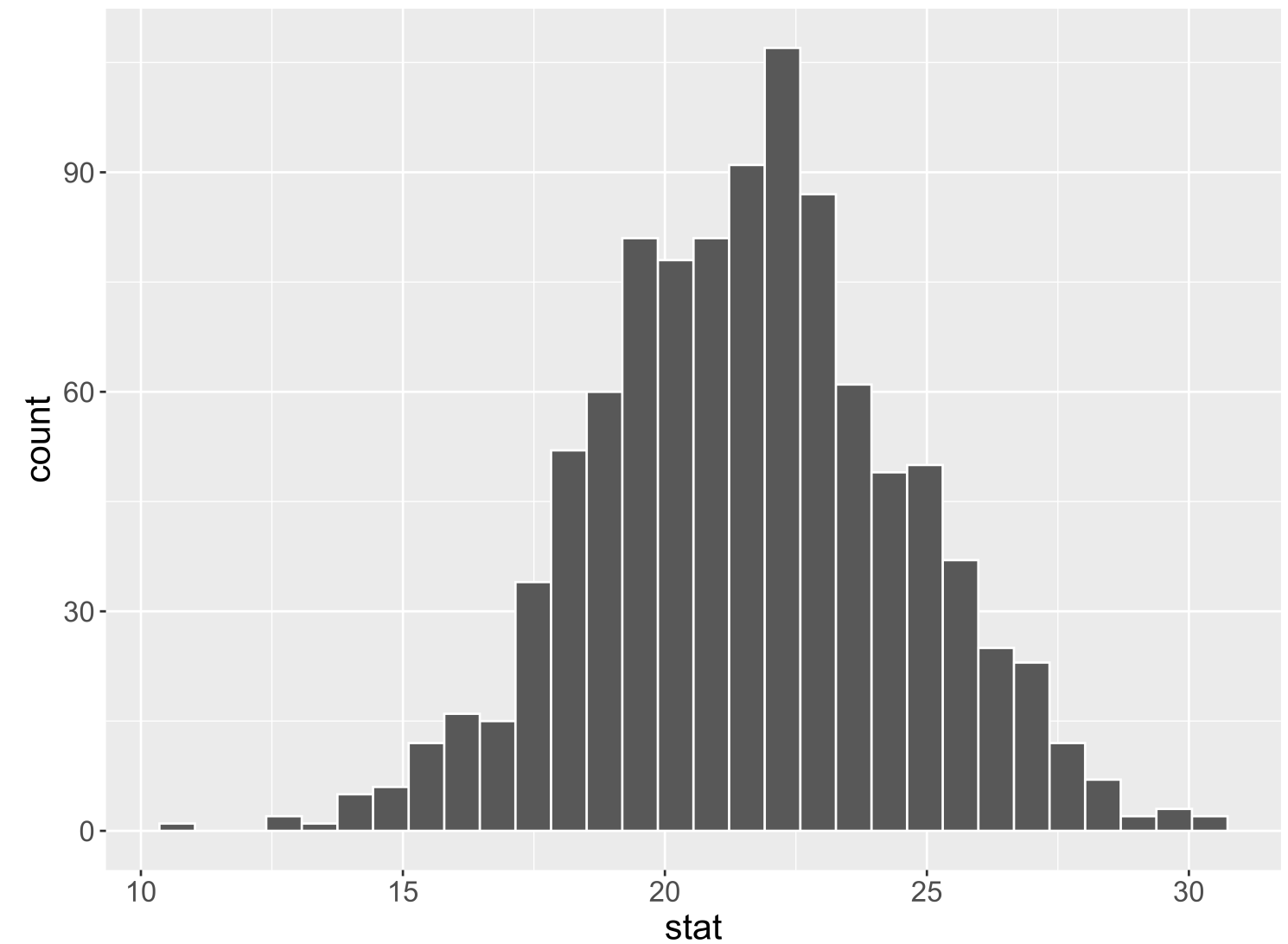
```
1 # Compute the summary statistic
2 diff_x_bar <- movies %>%
3   drop_na(OpeningWeekend) %>%
4   filter(Genre %in% c("Drama", "Action")) %>%
5   specify(OpeningWeekend ~ Genre) %>%
6   calculate(stat = "diff in means")
7 diff_x_bar
```

```
Response: OpeningWeekend (numeric)
Explanatory: Genre (factor)
# A tibble: 1 × 1
  stat
  <dbl>
1  21.7
```

- Why a difference in means?

Estimation for Difference in Means

```
1 # Construct bootstrap distribution
2 bootstrap_dist <- movies %>%
3   drop_na(OpeningWeekend) %>%
4   filter(Genre %in% c("Drama", "Action")) %>%
5   specify(OpeningWeekend ~ Genre) %>%
6   generate(reps = 1000, type = "bootstrap") %>%
7   calculate(stat = "diff in means",
8             order = c("Action", "Drama"))
9
10 # Look at bootstrap distribution
11 ggplot(data = bootstrap_dist,
12        mapping = aes(x = stat)) +
13   geom_histogram(color = "white")
```



Estimation for Difference in Means – SE Method

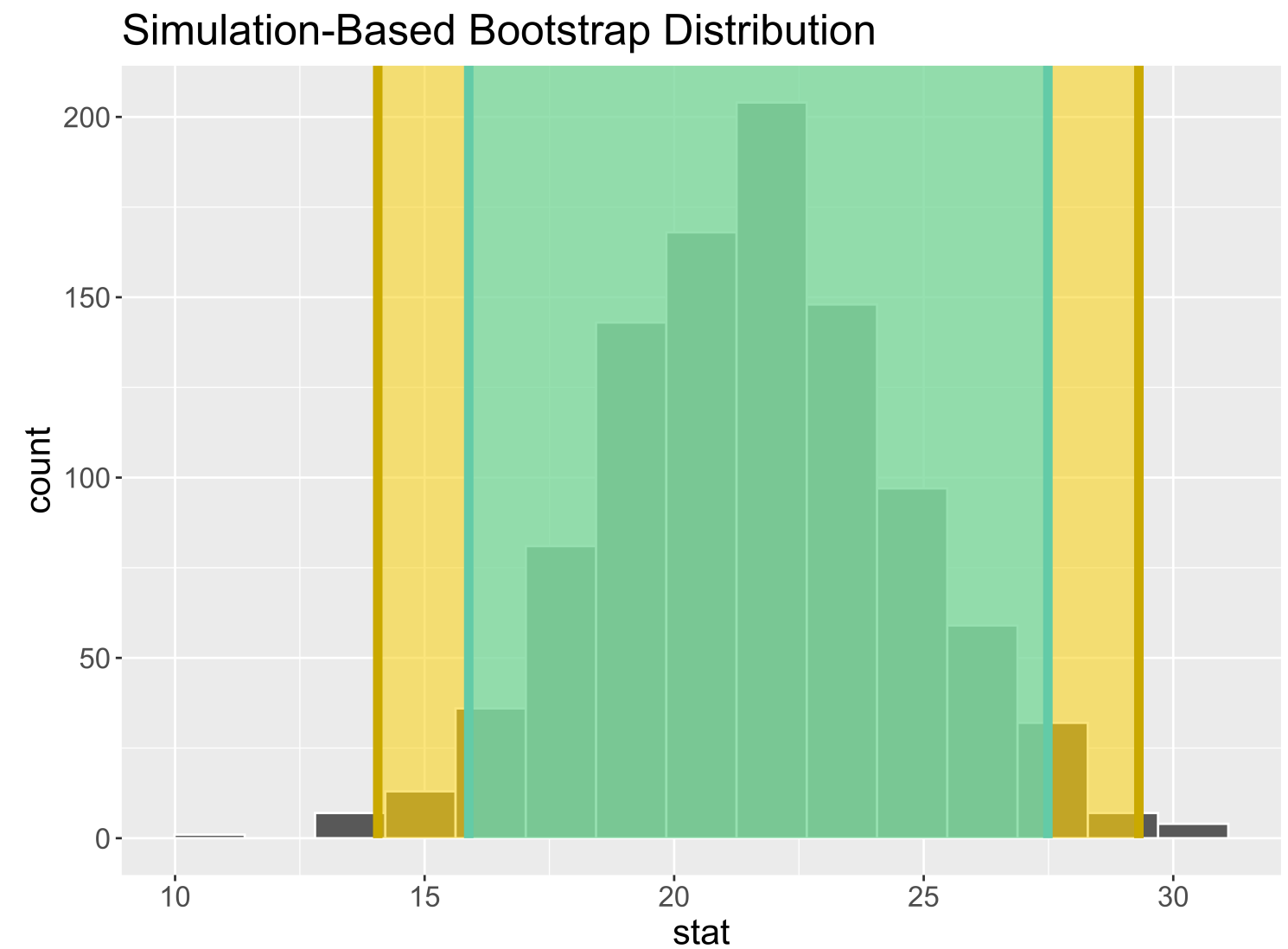
```
1 # Get confidence interval
2 ci_95 <- bootstrap_dist %>%
3   get_confidence_interval(type = "se", level = 0.95,
4                           point_estimate = diff_x_bar)
5 ci_95
```

```
# A tibble: 1 × 2
  lower_ci upper_ci
  <dbl>    <dbl>
1    15.9     27.5
```

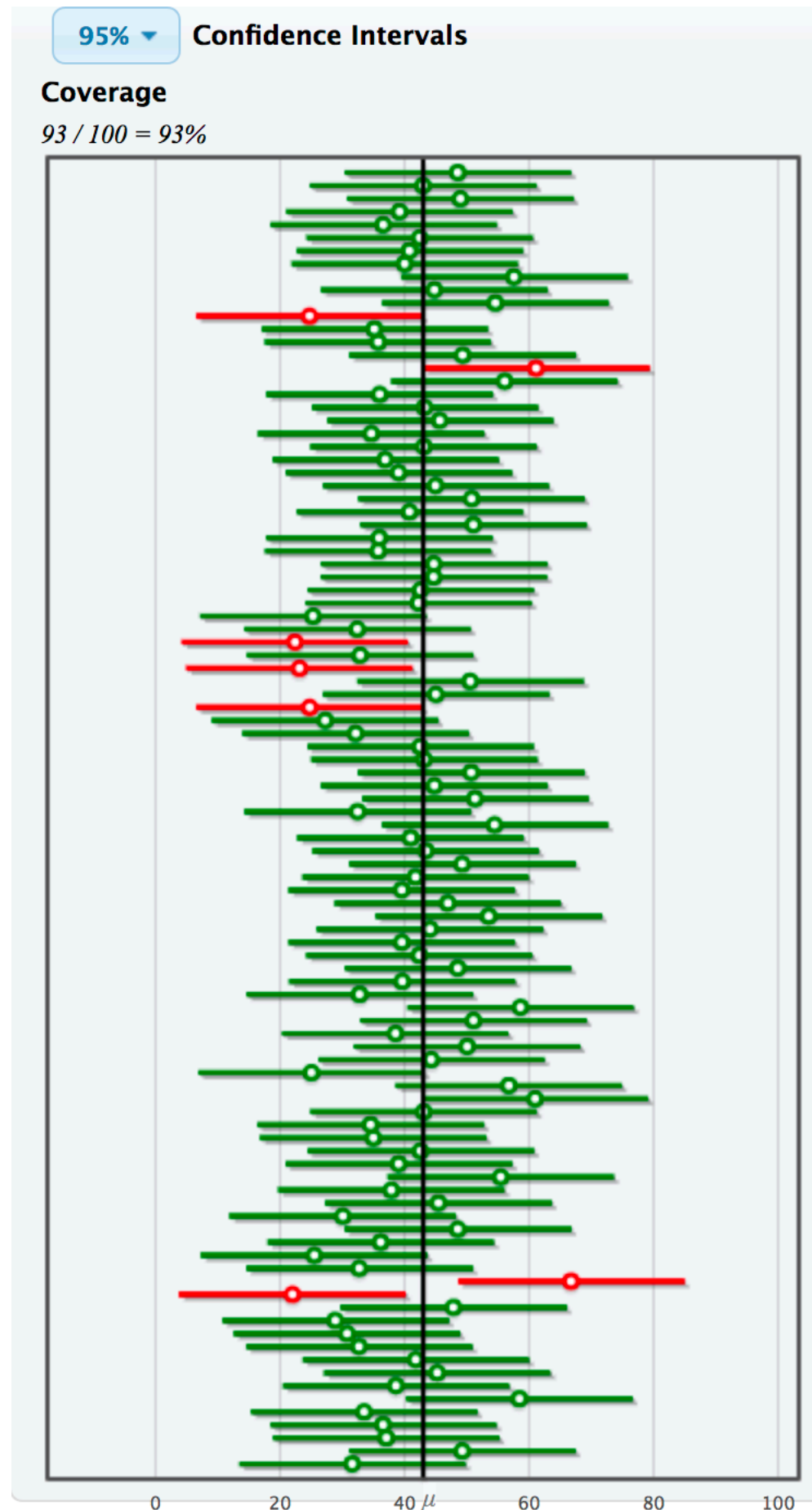
Interpretation: The point estimate is \$ 21.7M. I am 95% confidence that action movies make, on average, between \$ 15.9M and \$ 27.5M more than dramas.

Comparing CIs

```
1 ci_99 <- bootstrap_dist %>%  
2   get_confidence_interval(type = "se", level = 0.99,  
3                           point_estimate = diff_x_bar)  
4  
5 bootstrap_dist %>%  
6   visualize() +  
7   shade_confidence_interval(endpoints = ci_99,  
8                             fill = "gold1",  
9                             color = "gold3") +  
10  shade_confidence_interval(endpoints = ci_95)
```



- Why construct a 95% CI versus a 99% CI?



What do we mean by confidence?

- Confidence level = success rate of the method under **repeated sampling**
- How do I know if my ONE CI successfully contains the true value of the parameter?
- As we increase the **confidence level**, what happens to the width of the interval?
- As we increase the **sample size**, what happens to the width of the interval?
- As we increase the **number of bootstrap samples** we take, what happens to the width of the interval?

Interpreting Confidence Intervals

Example: Estimating average household income before taxes in the US

SE Method Formula:

$$\text{statistic} \pm \text{ME}$$

```
# A tibble: 1 × 3  
  ME lower upper  
<dbl> <dbl> <dbl>  
1 1963. 60517. 64443.
```

“The margin of [sampling] error can be described as the ‘penalty’ in precision for not talking to everyone in a given population. It describes the range that an answer likely falls between if the survey had reached everyone in a population, instead of just a sample of that population.” – Courtney Kennedy, Director of Survey Research at Pew Research Center

CI = interval of **plausible** values for the **parameter**

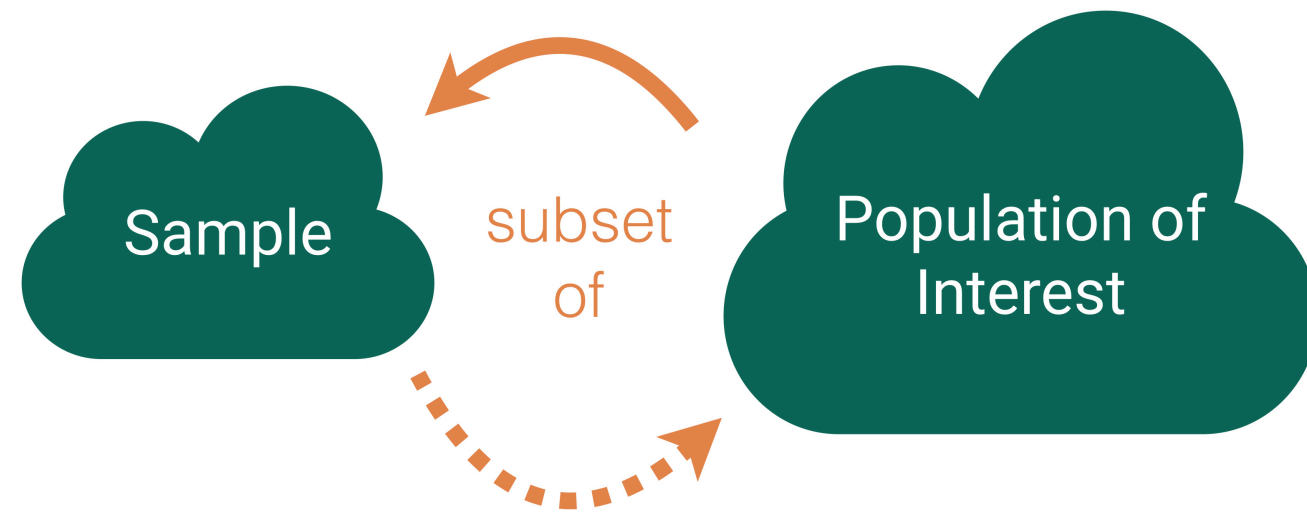
Safe interpretation: I am P% confident that {insert what the parameter represents in context} is between {insert lower bound} and {insert upper bound}.

Caution: Intervals in the wild

Statement in [an article](#) for The BMJ (British Medical Journal):

In many publications a $\pm$ sign is used to join the standard deviation (SD) or standard error (SE) to an observed mean—for example, 69.4 ± 9.3 kg. That notation gives no indication whether the second figure is the standard deviation or the standard error (or indeed something else). A review of 88 articles published in 2002 found that 12 (14%) failed to identify which measure of dispersion was reported (and three failed to report any measure of variability).⁴ The policy of the *BMJ* and many other journals is to remove $\pm$ signs and request authors to indicate clearly whether the standard deviation or standard error is being quoted. All journals should follow this practice.

Statistical Inference



Goal: Draw conclusions about the population based on a sample.

Main Flavors:

- Estimating numerical quantities.
- Testing conjectures.

Example: Does Extrasensory Perception (ESP) exist?



Daryl Bem and Ben Honorton

Bem and Honorton conducted extrasensory perception studies:

- A “sender” randomly chooses an object out of 4 possible objects and sends that information to a “receiver”.
- The “receiver” is then given a set of 4 possible objects and they must decide which one most resembles the object sent to them.

Out of 329 trials, the “receivers” correctly identified the object 106 times.

ESP Example

Let's consider the following questions:

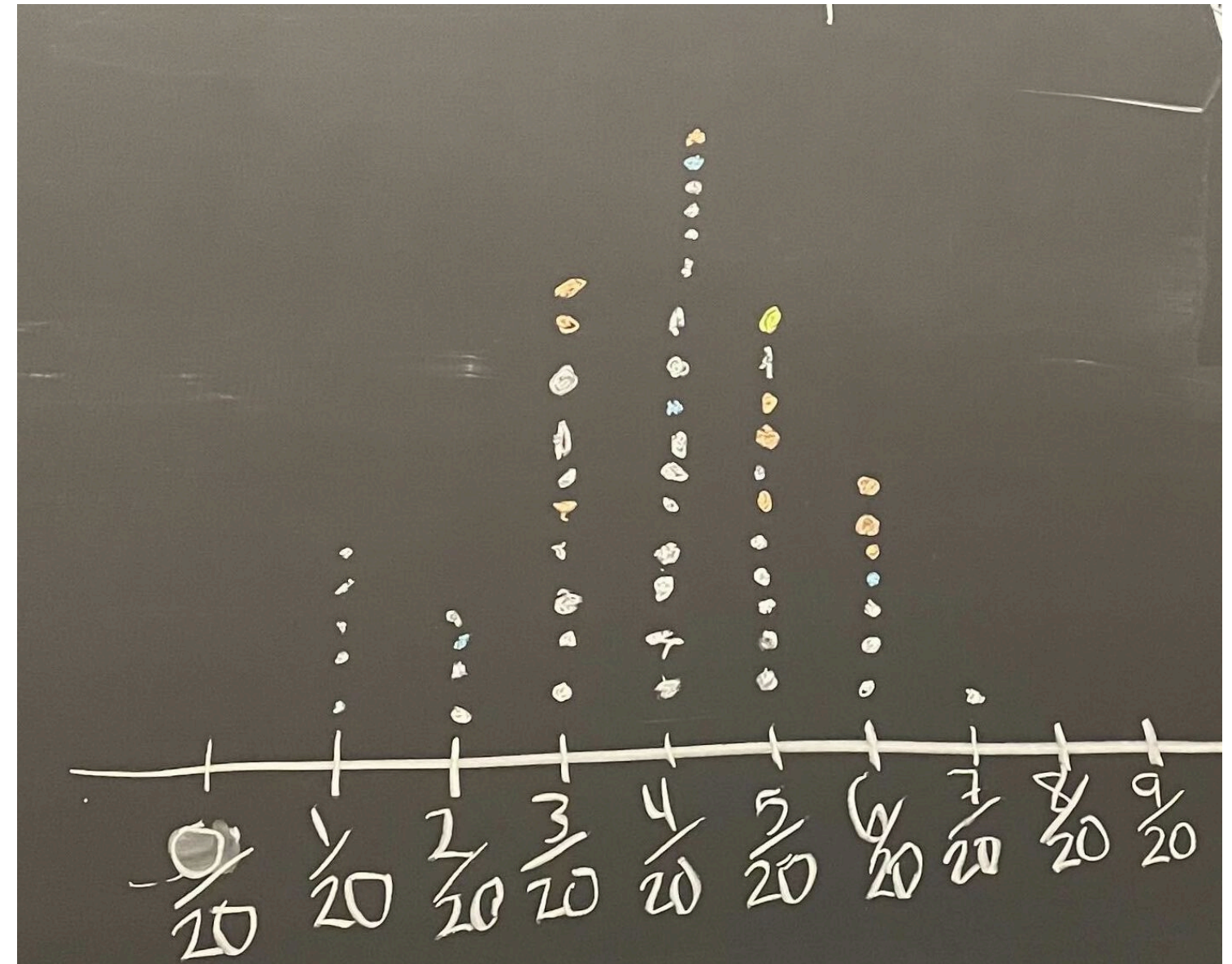
- a. If ESP does not exist and the “receivers” are guessing, how often would we expect them to be correct?
- b. For each sample (set of 329 trials), do we expect the proportion of correct guesses to be equal? Why or why not?
- c. Is it possible to randomly guess correctly 106 out of 329 times (i.e., 32% of the time)?
- d. How unusual is it to guess correctly 106 out of 329 times if ESP doesn't exist?

To help us answer d., we need a **sampling distribution** for the sample proportion where **we assume the “receivers” were purely guessing!**

Sampling Distribution of a Statistic

Steps for (Approximate) Distribution:

1. Decide on a sample size, n .
2. Randomly select a sample of size n from the population.
3. Compute the sample statistic.
4. Put the sample back in.
5. Repeat Steps (2) - (4) many (1000+) times.



Sampling Distribution Under No ESP

Steps for (Approximate) Distribution:

1. Decide on a sample size, n .
2. Randomly select a sample of size n from the population.

```
1 library(mosaic)
2 rflip(n = 329, prob = 0.25)
```

Flipping 329 coins [Prob(Heads) = 0.25] ...

```
T T H T H T T T H H T H H T T T T H H T H H H T H H H H T T H T H T T T
T H T T T T T T T T T H T T H T H H T T H T T H H H T T T T T T T H
H T T T H T T H H H T T T T T T T H T T T T H H T T T H T T T H T
T H T H T T H H T T T H T T T H T T T T H T T T H T H H H T H T T T T
H T T T T T T T T H H T T T T T H H T T T T H T T T H T T T H T T
T H T T H T H T T T T H T T T H T T T H T H T T H H H T H T T T T T
T T T T T H T T T T H T T T T T T T T T T H T T T H T T T H T H
T T T H T T T T T T T T T T H T H T T T T T T T T H T T H H T T
T T T T H T T H T T H T H T T T T T T T T T H T T T T T T T T
T H T T T
```

Number of Heads: 87 [Proportion Heads: 0.264437689969605]

Sampling Distribution Under No ESP

3. Compute the sample statistic.

```
1 rflip(n = 329, prob = 0.25, summarize = TRUE)
```

```
      n heads tails prob  
1 329     84   245 0.25
```

4. Put the sample back in.

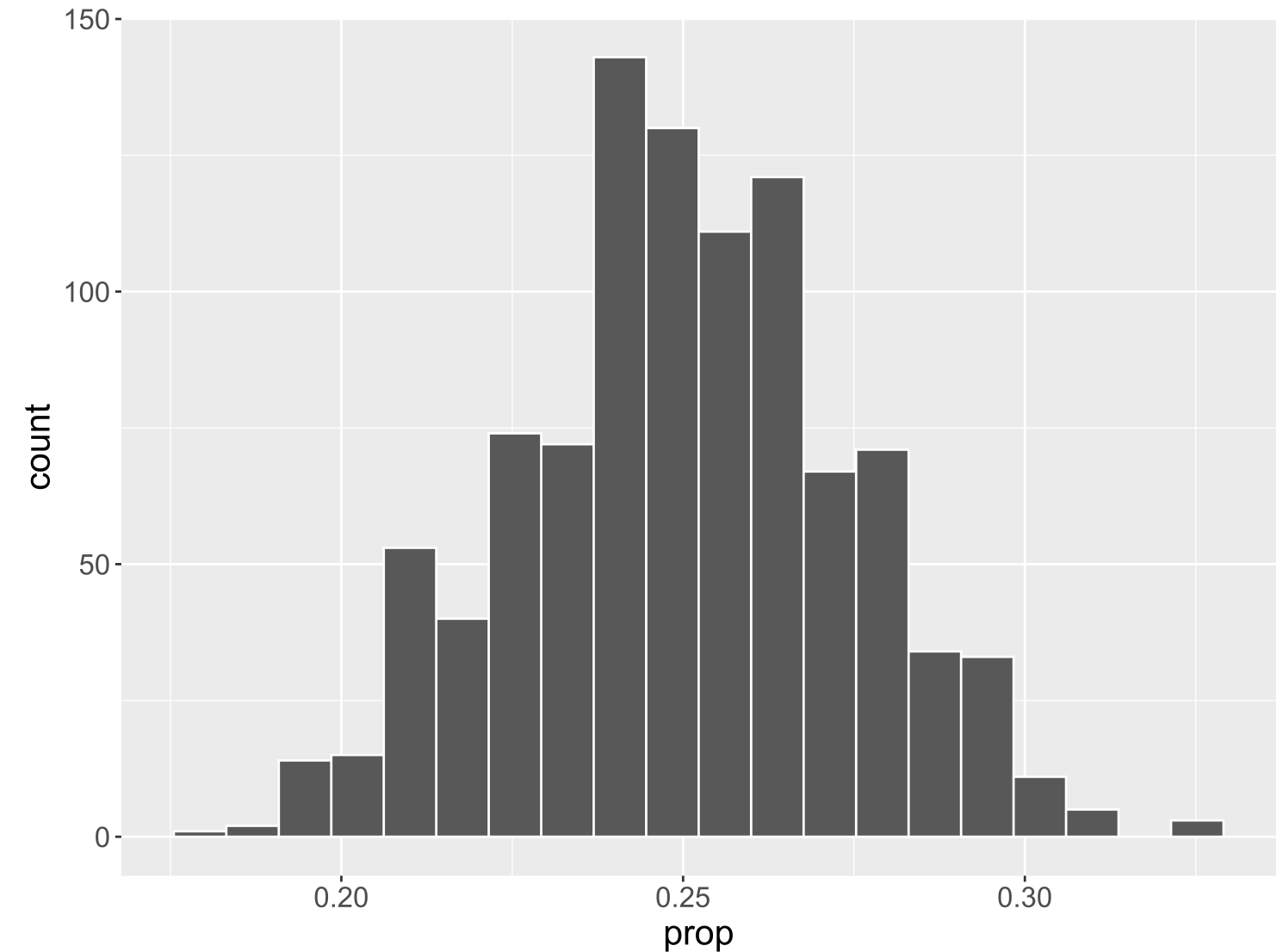
5. Repeat Steps (2) - (4) many (1000+) times.

```
1 guess_sampling_dist <- do(1000)*rflip(n = 329, prob = 0.25)  
2 guess_sampling_dist
```

```
      n heads tails      prop  
1    329     69   260 0.2097264  
2    329     71   258 0.2158055  
3    329     87   242 0.2644377  
4    329     79   250 0.2401216  
5    329     88   241 0.2674772  
6    329     86   243 0.2613982  
7    329     81   248 0.2462006  
8    329     69   260 0.2097264  
9    329     77   252 0.2340426  
10   329     75   254 0.2279635  
11   329     89   240 0.2705167  
12   329     84   245 0.2553191  
13   329     85   244 0.2583587  
14   329     81   248 0.2462006  
15   329     87   242 0.2644377
```

Sampling Distribution Under No ESP

```
1 ggplot(data = guess_sampling_dist,  
2       mapping = aes(x = prop)) +  
3   geom_histogram(color = "white",  
4                 bins = 20)
```

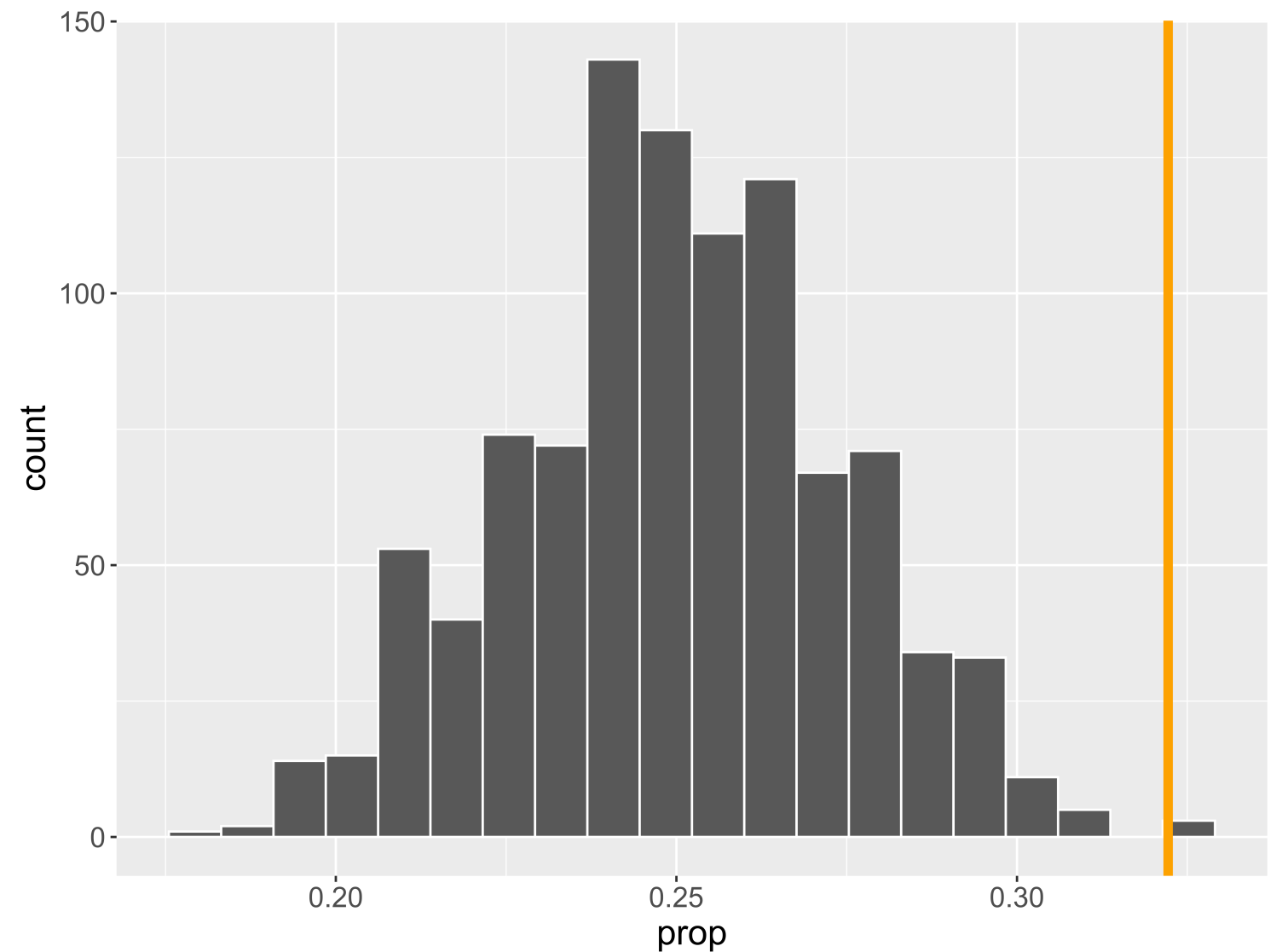


What value should our sampling distribution be centered around if the receivers are just guessing?

Sampling Distribution Under No ESP

- How do the study results compare to the sampling distribution under no ESP?
 - How unusual is it to guess correctly 106 out of 329 times if ESP doesn't exist?

```
1 p_hat <- 106/329
2 ggplot(data = guess_sampling_dist,
3       mapping = aes(x = prop)) +
4   geom_histogram(color = "white",
5                 bins = 20) +
6   geom_vline(xintercept = p_hat,
7             color = "orange",
8             size = 2)
```

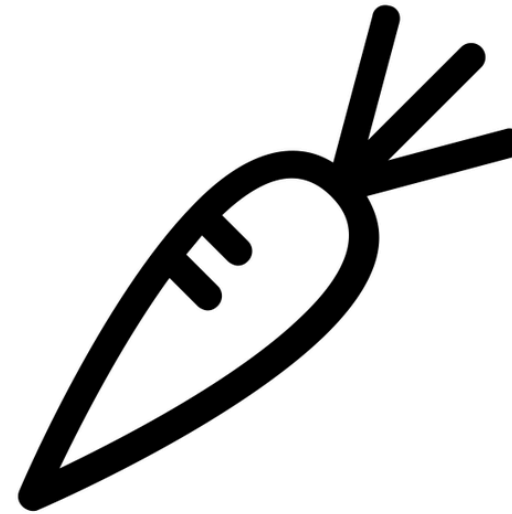


- Do Bem and Honorton have evidence that ESP exists?

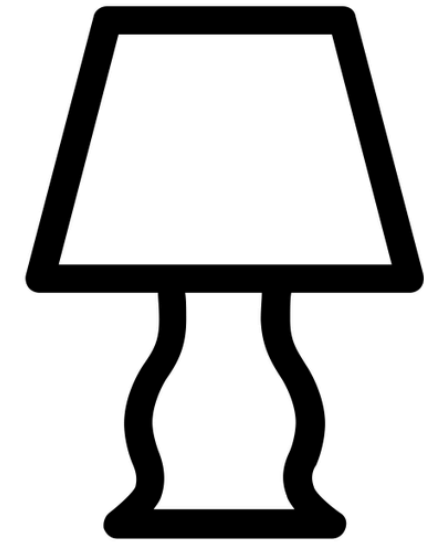
Do Bucknellians Have ESP?

In pairs:

- Decide who is going to be the sender and who is going to be the receiver.
- **Sender:** Think of one of these images.
- **Receiver:** Guess which image the sender was thinking of.
- Now switch roles and do it again!
- Once you have both played each role, each person should add a tally mark on the chalkboard.



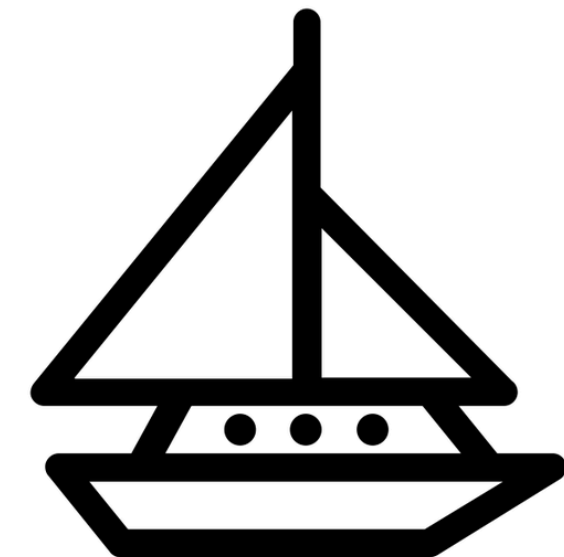
Created by Focus Lab
from Noun Project



Created by Focus Lab
from Noun Project



Created by Focus Lab
from Noun Project



Created by Focus Lab
from Noun Project

Do Bucknellians Have ESP?

What do we need to modify in the code to answer the question?

```
1 guess_sampling_dist <- do(1000)*rflip(n = 119, prob = 0.25)
2 p_hat <- 42/119
3 ggplot(data = guess_sampling_dist, mapping = aes(x = prop)) +
4   geom_histogram(color = "white", bins = 20) +
5   geom_vline(xintercept = p_hat, color = "orange", size = 2)
```

Hypothesis Testing

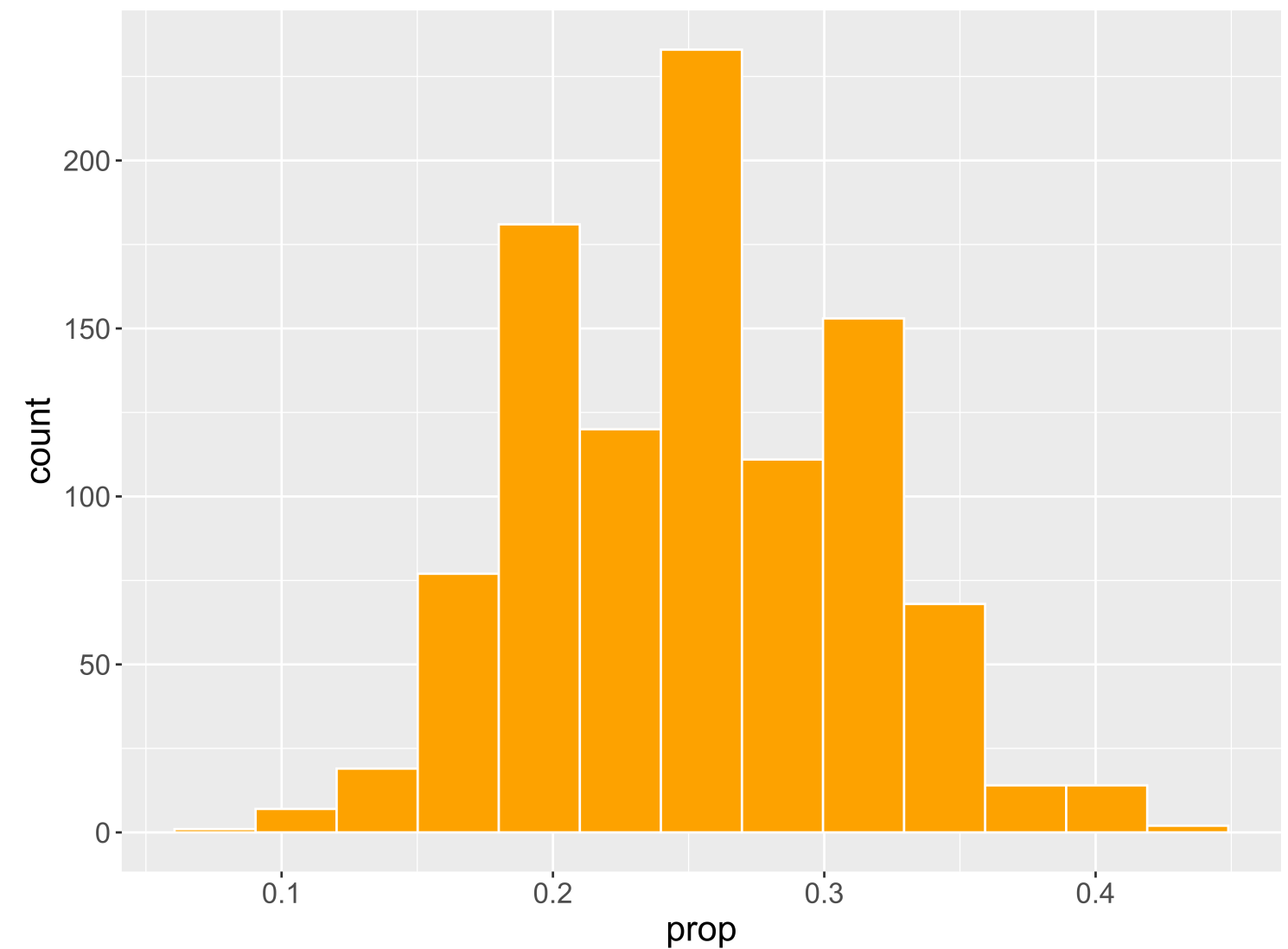
Big Idea:

- Make an assumption about the population **parameter**.
- Generate a sampling distribution for a *test* statistic based on that assumption.
 - Called a **null distribution**
- See if the test statistic based on the observed sample aligns with the generated sampling distribution or not.
- If it does, then we didn't learn much.
 - (Didn't prove the parameter equals the assumed value but it is still plausible)
- If it doesn't, then we have evidence that our assumption about the parameter was wrong.

ESP Example

Big Idea:

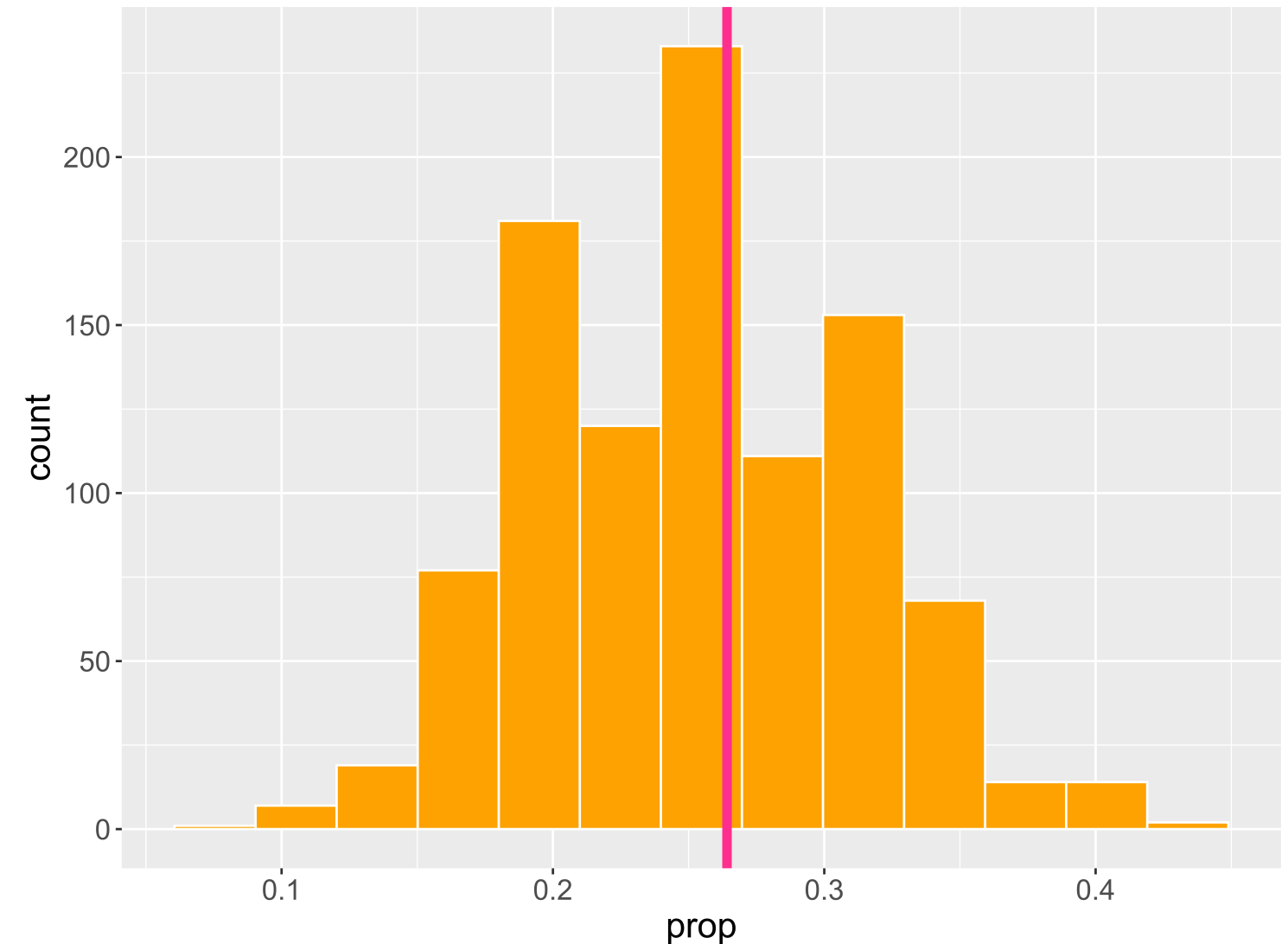
- Make an assumption about the population parameter.
 - Ex: ESP doesn't exist. p , probability of guessing correctly, equals 0.25.
- Generate a sampling distribution for a *test* statistic based on that assumption.
 - Called a **null distribution**



ESP Example

Big Idea:

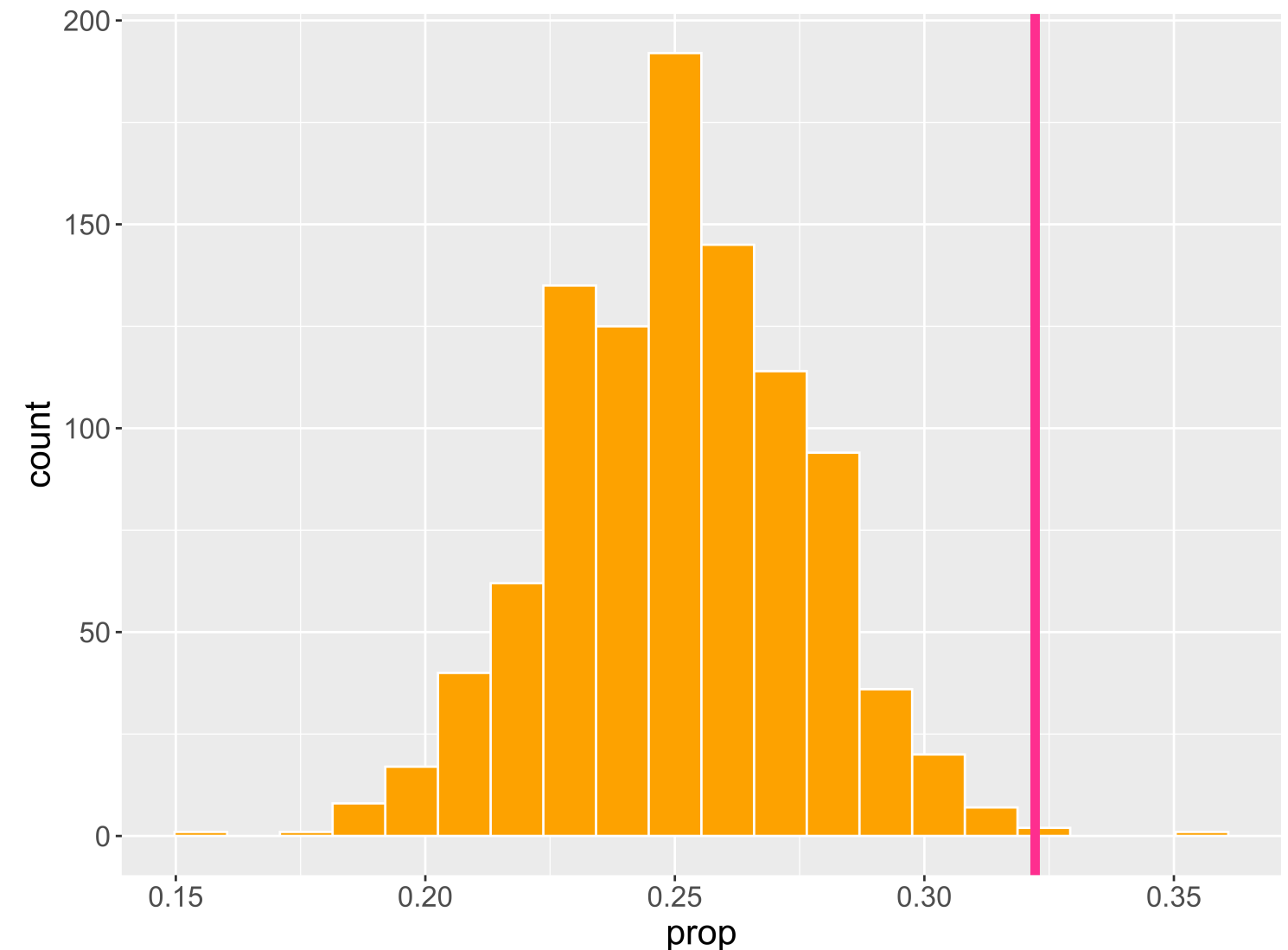
- See if the test statistic based on the observed sample aligns with the generated sampling distribution or not.
 - Ex: It is in the center-ish of the distribution. It isn't an unusual value.
- If it does, then we didn't learn much. (Didn't prove the parameter equals the assumed value but it is still plausible)
 - It is still possible that ESP doesn't exist.



ESP Example

Big Idea:

- See if the test statistic based on the observed sample aligns with the generated sampling distribution or not.
 - It is far in the tails of the distribution. It is an unusual value.
- If it doesn't, then we have evidence that our assumption about the parameter was wrong.
 - We have evidence ESP exists.



Let's Take a Step Back from Our Last Statement...

- Two important words in data analysis:
 - Reproducibility
 - Replicability
- **Reproducibility**: If I give you the raw data and my write-up, you will get to the exact same final numbers that I did.
- By using **Quarto** Documents, we are learning a **reproducible** workflow.
- **Replicability**: If you follow my study design but collect new data (i.e. repeat my study on new subjects), you will come to the same conclusions that I did.

Replication Crisis

- Science is going through a **replication crisis** right now.
 - In cancer science, many “discoveries” don’t hold up
 - Estimating the reproducibility of psychological science
 - Psychology Is Starting To Deal With Its Replication Problem
- And, sadly, **replication** studies of Bem and Honorton’s ESP trials typically failed to find evidence of ESP.

Reminders:

- Map making in **R** workshop: Part of the Sustainability Symposium
 - Th, March 26, 2:30 - 3:50pm, Taylor 210

